MUHAMAD SOLEHUDDIN

Manufacturing Automation Engineer
Penang, Malaysia | +60 19 228 7350 | saladiniart@gmail.com
Portfolio: salbotics.uk/projects | GitHub: github.com/SaladinIART
LinkedIn: linkedin.com/in/solehuddin-muhamad-b67068132

PROFESSIONAL SUMMARY

Manufacturing automation engineer and Graduate Technologist with hands-on experience in production-equipment maintenance, testing, CAD/CAM and factory digitization. Connects equipment signals and industrial communications (Modbus, RS-485, OPC UA, MQTT) to data collection, SQL storage and dashboards, including a factory energy-monitoring system deployed in a live manufacturing environment. Builds independently documented automation studies with AI-assisted development, verified through automated tests, CI and independent review, and states clearly what is deployed, simulated or not yet proven.

CORE SKILLS

- Equipment support: preventive maintenance, fault isolation, sensor and PLC input checks, wiring and communication checks, leak testing, retesting and technical handover.
- Industrial integration: Modbus RTU/TCP, RS-485, OPC UA, MQTT, edge data acquisition, Node-RED, TimescaleDB and Grafana.
- Software and controls: Python and SQL project work with AI-assisted development; FastAPI, Git, Docker Compose, automated software tests and PLC sequence simulation.
- Mechanical and manufacturing: SolidWorks, Mastercam, CAD/CAM, CNC troubleshooting, electrical and mechanical drawing interpretation.
- Engineering documentation: requirements, test specifications, technical reports, condition assessment, failure and recovery evidence, and documented limitations.

PROFESSIONAL EXPERIENCE

Test Technician

Electronics manufacturing, Penang | Jul 2026 - Present

- Perform procedure-led PLC input and sensor checks; investigate abnormal indications through wiring, installation, power and connection checks.
- Support controlled test setup and computer-to-PLC communication checks; conduct vacuum/helium leak tests, record findings and support retesting after corrective work.

Freelance Systems Engineer

Salbotics Solutions, Penang | Oct 2025 - Aug 2026

- Built and documented self-directed industrial software demonstrations with AI assistance, recording requirements, test scenarios and implementation limits; these are portfolio projects, not client deployments.
- Researched prospective client requirements and prepared engineering proposals; implementation remained subject to client agreement.

System Engineer, Digitization Department

Alumac Industries Sdn. Bhd., Sungai Buloh | Jan 2024 - Sep 2025

- Developed and supported factory energy monitoring, integrating industrial measurements, Modbus RTU/RS-485 communications, edge collection, SQL storage and Node-RED dashboards.
- Diagnosed faults across equipment signals, serial communications, collection services, databases and dashboards in a live manufacturing environment.
- Supported monitoring integration for existing production equipment and documented operator and maintenance handover needs.
- Supported manufacturing reporting and Microsoft 365 workflows using Power BI, SharePoint, Teams, Power Automate and Power Apps.

Maintenance Technician

Diversatech (M) Sdn. Bhd., Klang | May 2021 - Dec 2023

- Performed preventive maintenance, repairs and structured fault isolation on production equipment; documented equipment condition, corrective work and issues requiring escalation.

CAD and CAM Technician

AMS Seihin Sdn. Bhd., Kajang | Jun 2016 - Sep 2018

- Used Mastercam for CNC programming and troubleshooting on Brother and HAAS machining centres with shop-floor verification; supported SolidWorks design, drawing interpretation and mechanical work.

SELECTED ENGINEERING PROJECTS

Full case studies with objective, decision and result: salbotics.uk/projects

Factory Energy Visibility - Deployed

salbotics.uk/projects/factory-energy-monitor

- Brought machine-level energy readings from industrial meters over Modbus RTU/RS-485 into edge services, historical SQL storage and operator dashboards in a live manufacturing environment; public case study is anonymized.

NEXUS OPC UA and IIoT Condition Assessment - Validated demo

salbotics.uk/projects/nexus-iiot | github.com/SaladinIART/IIoT-Telemetry-Stack

- Selected and directed AI-assisted implementation of a read-only OPC UA extension connecting a simulated furnace to MQTT, TimescaleDB, FastAPI and Grafana, retaining the existing Modbus path.
- Separated equipment state from data quality across steady, frozen, unreliable-source, process-fault and connection-recovery cases. Evidence: 62 passing automated cases (53 unit, 9 integration) and a CI trace matching an acknowledged sample to its exact database row and Grafana value.

BocoBot Condition Assessment and Warehouse Routing Study - Simulated study

salbotics.uk/projects/bocobot-condition-routing

- Framed and directed an AI-assisted study of synthetic shifted, noisy and incomplete line-sensor observations across 30 runs in six scenarios, documenting the limits of physical-cause inference.
- Compared Dijkstra and A* routing on an invented warehouse graph; routing passed 20 tests and independent review checked path costs across all 256 ordered node pairs.

PLC Sequence Verification and Asset Vision Checks - Simulated study

salbotics.uk/projects/plc-asset-vision-checks

- Checked four PLC training-model traces (normal timing, conflicting START/STOP and reset, stop/restart, emergency/restart); all 18 declared checks matched.
- Checked Asset Vision QR decoding on five clean sample frames; all five payloads matched. No physical PLC, camera or object-detection claim.

AI-Assisted Engineering Workflow - Working practice

salbotics.uk/projects/ai-assisted-engineering-workflow

- Coordinate AI-assisted engineering through bounded briefs, evidence requirements and cross-model review, retaining failed checks and corrections before accepting a result.

EDUCATION AND PROFESSIONAL REGISTRATION

Bachelor of Engineering Technology (Automation and Robotics) with Honours

UniKL Malaysia France Institute | 2017 - 2023

Diploma in Industrial Electronics (Mechatronics)

German-Malaysian Institute | 2014 - 2017

Graduate Technologist (GT), Manufacturing and Industrial Technology

Malaysia Board of Technologists (MBOT) | 2026

SELECTED PROFESSIONAL DEVELOPMENT

- Become a PLC Developer | LinkedIn Learning | Dec 2025
- Continuous Deployment Strategies by Pearson | LinkedIn Learning | Mar 2026
- EFK Stack - Enterprise-Grade Logging and Monitoring | LinkedIn Learning | May 2026